

LESSON 7-5 • SESSION 1 • PRACTICE • FORM B**Plotting Points on the Coordinate Plane**

Learning target – I can plot a point with rational coordinates and name the quadrant it lies in. · 6.NS.C.6c

REMEMBER

- An **ordered pair** (x, y) names a point. Start at the **origin** $(0, 0)$: x says right or left, then y says up or down.
- Signs name the **quadrant**: $(+, +)$ I, $(-, +)$ II, $(-, -)$ III, $(+, -)$ IV.
- A point with a 0 coordinate is **on an axis**, not in a quadrant.

WORKED EXAMPLEPlot $(-5, -4)$. Name its quadrant.**Step 1** Start at the origin, $(0, 0)$.**Step 2** Move 5 units left, because $x = -5$.**Step 3** Move 4 units down, because $y = -4$.**Answer** $(-, -)$ is Quadrant III

WORDS coordinate plane (*plano de coordenadas*) · ordered pair (*par ordenado*) · origin (*origen*) · quadrant (*cuadrante*)

PRACTICE

show your work · label units

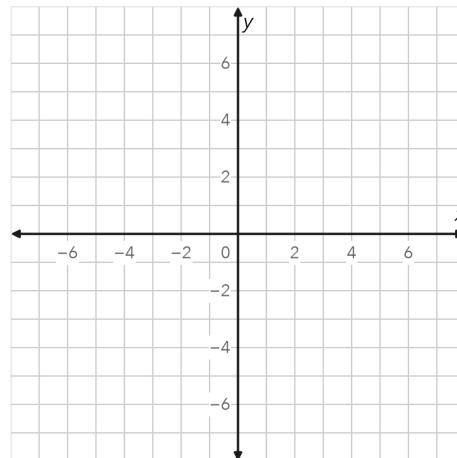
1. Name the quadrant or axis for each point.

a. $(-3, 8)$ _____b. $(5, 2)$ _____c. $(-6, -1)$ _____d. $(7, -4)$ _____e. $(0, 3)$ _____f. $(-6, 0)$ _____

2. **Park Map.** Ms. Rivera's class map puts the fountain at the origin. Gate $(3, 6)$, Pond $(-8, 4)$, Bench $(-2, -7)$, Swing Set $(6, -5)$.

Hint Move along the x -axis first.

- a. Plot and label all four spots.
- b. Quadrant: Gate ____ Pond ____ Bench ____ Swing Set ____
- c. Ordered pair for the fountain: _____



3. **Build the Pair.** A point is 9 units left of the y -axis and 5 units below the x -axis. Write its ordered pair and quadrant.

Hint Below means y is negative.**Point** _____

4. **Order Matters.** Name the quadrant of $(-4, 2)$ and of $(2, -4)$. Are they the same point?

Hint Check the sign of x , then the sign of y .**Quadrants** _____

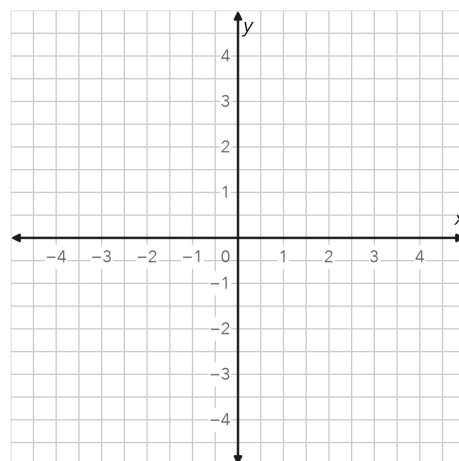
5. **Same Digits.** Theo says $(3, -7)$ and $(-7, 3)$ name the same point because they use the same two numbers. Explain why he is wrong.

6. **Rational Coordinates.** Plot $(-2.5, -4.5)$ and $(3.5, 1.5)$.

Name each quadrant.

Hint Count 2 whole units, then half of the next.

Quadrants _____



7. **City Map.** The library is at the origin. The museum is at $(6, -3)$. The stadium is at $(-8, 2)$.

a. Quadrant: museum ____ stadium ____

b. Ordered pair for the library: _____

8. **Map Points.** A map shows a school at $(3.5, -2)$, a park at $(-3.5, -2)$, and a pool at $(-3.5, 4)$.

a. Name the quadrant of each place. _____

b. Which y-coordinate do two places share? _____

c. How many units straight up is the pool from the park? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Jalen's Work.** Jalen plotted $(-3, 6)$ by moving down 3 units and then right 6 units. His point landed in Quadrant IV. Explain his mistake. Then describe the correct moves and quadrant.

TEST PRACTICE

choose the best answer

10. The point $(5, -2)$ lies in –

A Quadrant I

B Quadrant II

C Quadrant III

D Quadrant IV

11. A point with both coordinates negative lies in –

A Quadrant I

B Quadrant II

C Quadrant III

D Quadrant IV

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

(9, 5) Z	(-9, -5) I	II P	(-5, -9) Q	(0, 0) N	y-axis O	2 X	-2 T	6 S
#1a		#1e	#3	#7b	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____